Practical 5 - Understanding Two Tailed Test

Problem statement - Dataset to be used is: ratings

Q1. Using the teachers' rating dataset, what is the probability of receiving an evaluation score of greater than 4.5

Q2. Using the teachers' rating dataset, what is the probability of receiving an evaluation score greater than 3.5 and less than 4.2

Q3. Using the two-tailed test from a normal distribution:

1. A professional basketball team wants to compare its performance with that of players in a regional league.

2. The pros are known to have a historic mean of 12 points per game with a standard deviation of 5.5.

3. A group of 36 regional players recorded on average 10.7 points per game.

4. The pro coach would like to know whether his professional team scores on average are different from that of the regional players.

State the null hypothesis Null Hypo: ("The mean point of the regional players is not different from the historic mean") Alternative Hypo: ("The mean point of the regional players is different from the historic mean")

Solution- Here is the complete code

Import Libraries

```
import pandas as pd
import numpy as np
import math
from scipy.stats import norm
```

Load Dataset

```
df = pd.read_csv('/content/teachingratings_extracted.csv')
```

Q1. Using the teachers' rating dataset, what is the probability of receiving an evaluation score of greater than 4.5

```
total_teachers = len(df)
count_greater_4_5 = (df["eval"] > 4.5).sum()
probability_q1 = count_greater_4_5 / total_teachers

print("Q1: Probability of eval > 4.5")
print("Count =", count_greater_4_5)
print("Total =", total_teachers)
print("Probability =", probability_q1)
print("Percentage =", probability_q1 * 100, "%")
```

```
Q1: Probability of eval > 4.5
Count = 78
Total = 458
Probability = 0.1703056768558952
Percentage = 17.03056768558952 %
```

Q2. Using the teachers' rating dataset, what is the probability of receiving an evaluation score greater than 3.5 and less than 4.2

```
count_between = ((df["eval"] > 3.5) & (df["eval"] < 4.2)).sum()
probability_q2 = count_between / total_teachers

print("\nQ2: Probability of 3.5 < eval < 4.2")
print("Count =", count_between)
```

```
print("Total =", total_teachers)
print("Probability =", probability_q2)
print("Percentage =", probability_q2 * 100, "%")
```

```
Q2: Probability of 3.5 < eval < 4.2
Count = 195
Total = 458
Probability = 0.425764192139738
Percentage = 42.5764192139738 %
```

Q3. Using the two-tailed test from a normal distribution:

1. A professional basketball team wants to compare its performance with that of players in a regional league.
2. The pros are known to have a historic mean of 12 points per game with a standard deviation of 5.5.
3. A group of 36 regional players recorded on average 10.7 points per game.
4. The pro coach would like to know whether his professional team scores on average are different from that of the regional players. State the null hypothesis Null Hypo: ("The mean point of the regional players is not different from the historic mean") Alternative Hypo: ("The mean point of the regional players is different from the historic mean")

```
# Given values
mu0 = 12             # Historic mean
sigma = 5.5          # Population standard deviation
n = 36               # Sample size
sample_mean = 10.7   # Sample mean
alpha = 0.05         # Significance level

# Step 1: Calculate Standard Error
SE = sigma / math.sqrt(n)

# Step 2: Calculate Z-test statistic
Z = (sample_mean - mu0) / SE

# Step 3: Calculate two-tailed p-value
p_value = 2 * norm.cdf(-abs(Z))

print("Standard Error =", SE)
print("Z-test statistic =", Z)
print("Two-tailed p-value =", p_value)
```

```
if p_value < alpha:
    print("Reject H0")
    print("Conclusion: The mean is significantly different from 12.")
else:
    print("Fail to reject H0")
    print("Conclusion: There is not enough evidence to conclude that the mean is different from 12.")
```

```
Standard Error = 0.9166666666666666
Z-test statistic = -1.418181818181819
Two-tailed p-value = 0.15613768876786
Fail to reject H0
Conclusion: There is not enough evidence to conclude that the mean is different from 12.
```

```
Start coding or generate with AI.
```